Minimal Test Set Generation for Input Stuck-at and Bridging Faults in Reversible Circuits

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***Abstract*—Research in reversible computing has gained importance because of its potential use in low-power design, and also quantum computing. Several works on reversible circuit testing have also been reported. Many fault models have been proposed, some of which have been borrowed from traditional logic. In this paper, we consider the problem of reversible circuit testing, specifically targeting test generation for stuck-at and bridging fault models. It has been shown that $\lceil \log_2 N \rceil$ number of test vectors are necessary and sufficient for detection of all inputs bridging faults of a reversible circuit with N inputs. Addition of only one more test vector to an existing test set of single input bridging faults can also detect all single input stuck-at faults. Finally, we provide our experimental results to verify our claim, and also show that the test size is smaller as compared to existing methods.**

Keywords*—Reversible logic; Fault model; NCT library; GT library; Test set.

I. Introduction

Information is the necessary part of every computing system. In conventional digital circuits, number of input and output lines can be different, which typically results in loss of information and hence some mandatory power dissipation [1]. It has also been postulated [2] that if a circuit has to consume zero power, it must be reversible. This has motivated researchers to explore reversible logic as a circuit design alternative. Reversible circuits implement a bijective mapping, which requires the number of input and output lines to be equal. This property allows us to deduce the inputs from the outputs [3].

In the circuit design domain, the presence of manufacturing faults is a major concern that may lead to physical failure of the system. Even a single fault can cause large deviation in the expected performance of a system. A fault model is generally used to abstract the effects of physical failures and also helps to simplify the complexity reducing the number of faults [4]. Various testing approaches for reversible logic circuits have being proposed.

The generation of test patterns in a reversible circuit is much simpler than conventional logic circuits because the property of reversibility ensures high controllability and observability [5]. Because of this, backtracking that is considered to be difficult for conventional circuits is straightforward for reversible circuits. For every vector at the output of a reversible gate, there exists a unique vector at the input. A test set for detecting faults in a reversible circuit is said to be complete if all possible faults under some assumed fault model can be detected with the help of the test set. In [6], the authors discussed the connection between various fault models, particularly the single input stuck-at faults and the bridging faults. The bridging faults between two lines at the same level can be detected by a test vector that sets the two lines to opposite logic values, i.e., 01 or 10, which was reported in [4]. Stuck-at faults at any particular level also require lines to be set to 0 or 1 to detect the faults at that level.

Various works have been reported in the literature [6], [7], [8], [9] that address the generation of complete test sets based on some fault model. In this context, a key focus of the present work is deriving test patterns for a particular fault from the generated test patterns for another fault model. Specifically, we first generate the test patterns for single input bridging faults, and then subsequently reconstruct the test vectors for detecting single input stuck-at faults. The method produces a complete test set, which is also the minimal test set for detecting both single input bridging faults and single input stuck-at faults.

The paper is organized as follows: Section II introduces some basic background of reversible gates specially targeting the NOT, CNOT, Toffoli (NCT) and generalized (n-bit) Toffoli (GT) gate library and also provides an overview of the relevant fault models in reversible circuit. Section III describes our proposed method for generating complete and minimal test set. The experimental results are presented in Section IV. Finally, concluding remarks are presented in Section V with some possible directions for future works.

II. Background

A reversible logic circuit realizes an n-input, n-output function with no fan-out and feedback connections [10]. It can be implemented by a linear cascade of reversible gates [11]. Several reversible gates have been proposed in the literature, like controlled-NOT (CNOT) or Feynman [12], Toffoli [13], Fredkin [14], PRES [15], TR [16], SWAP [17], etc. The most commonly used gate library is the NCT library, consisting of NOT, CNOT and Toffoli gates as shown in Figure 1. The NOT gate is a single input gate that simply inverts the input. The CNOT gate passes the first input as it is and inverts the second input if the first input is 1. The Toffoli gate is a 3-input gate that inverts the third bit if the first two bits are 1. The GT library contains generalized multiple-control Toffoli gate. In this paper, we have used the NCT and GT gate libraries.

A. Fault Models in Reversible Circuit

Traditionally, various physical faults that occur in a system are modeled using some suitable fault models at different levels of abstraction like behavioral, functional, structural, geometric, etc. [4]. Fault models help to analyze faults and reduce the complexity of generating test vectors [6]. In this work, we

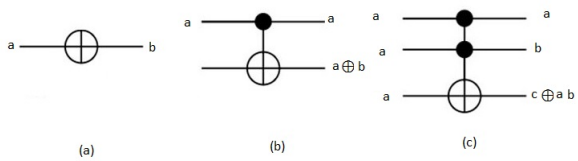

Fig. 1: (a) NOT (b) CNOT (c) Toffoli

consider the stuck-at and bridging fault models in reversible circuits.

1) Stuck-at Fault Model: A stuck-at fault causes some of the inputs or output lines of a gate to be fixed at either 0 (stuck-at-0) or 1 (stuck-at-1) [5], [18]. If only one stuck-at fault is considered at a time in the circuit, it is referred to as the single stuck-at fault (SSF) model. On the other hand, if multiple lines can have stuck-at faults at the same time, it is called multiple stuck-at fault (MSF) model. Patel et al. [5] proved that for a reversible circuit a test set for single stuck-at faults is complete if and only if each wire at every level can be set to both 0 and 1 by the test set.

2) Bridging Fault Model: A bridging fault is said to occur when two signal lines are connected together due to some fault. Depending on the logic circuitry employed, this may result in a wired-OR or wired-AND operation at the point of bridging [4]. Single bridging fault between two lines at the same level in a reversible circuit can be tested by an input vector if and only if the two affected lines are driven by opposite logic values [19].

B. Related Work

Some of the previous works that are relevant to the present paper are briefly reviewed in this section. In 2004, Patel et al.[5] proposed an Integer Linear Programming (ILP) based approach with binary decision variables to construct the minimal test set for detecting stuck-at faults as well as cell-faults in reversible circuits with gates from the NCT library. Because of its computational complexity, the method is applicable only for small circuits. To handle larger circuits, the authors suggested a circuit decomposition approach.In 2005, a method was proposed [20] for detecting all stuck-at faults that reduce the number of test vectors required to 3; however, an additional line and one extra control in every gate are required as overhead. In 2008, Ibrahim et al. [8] proposed a method for offline testing of multiple stuck-at faults in reversible circuits with only 2 test vectors by adding maximum two input lines. In this method some K-CNOT gates are replaced by (K+1)-CNOT gates and some are replaced by (K+2)-CNOT gates. In 2011, Nayeem et al. [9] proposed a technique to convert an ESOP-generated reversible circuit into an online testable circuit. Here one needs to add some CNOT gates and replace the Toffoli gates with the extended Toffoli gates (ETG). Overall, this method required double the number of CNOT gates.

In 2007, Bubna et al. [21] proposed a Design-For-Test (DFT) methodology for detecting bridging faults in reversible circuits. The proposed method required ($\lceil \log_2 N \rceil$)+3 test vectors for N input wires. Additional input wires are added to Toffoli gates to realize the DFT circuit. Rahaman et al. [19] showed that a test set of cardinality $(d \log_2 n)$ is sufficient to detect all intra-level bridging faults. In 2008, Sarkar et al. [22] presented a polynomial time algorithm for generating a set of universal test vectors for detecting all single and multiple input bridging faults of the reversible circuit.

It is observed from the literature that while trying to reduce the number of test vectors for complete fault coverage, there is an additional burden of extra circuitry in the form of additional input lines or control connections for reversible gates.

In this work, we have proposed a method to find the minimal set of test vectors to detect all possible input bridging faults without any additional hardware. Also we have shown that by adding one more test vector, we can also detect all possible input stuck-at faults.

III. Proposed Method

In this section, we first present a method to find the minimum set of test vector to detect all input bridging faults in a reversible circuit. Next we show that this test set can be enhanced by adding one more test vector to detect all the input stuck-at faults. Some definitions are presented, which are used in subsequent discussions.

Definition 1: A test vector (TV) is a set of binary inputs that is applied to a reversible circuit for the purpose of testing. Binary inputs $\langle b_1\ b_2\ \ldots\ b_N \rangle$ are provided to the input lines L = $\{L_1, L_2,\ \ldots,\ L_N\}$, where N is the number of input lines. Let the test set be TS = $\{TV_1,\ TV_2 ...,\ TV_l\}$, for $1 \leq j \leq l$, TV_j = $\langle b_{1j}\ b_{2j}\ \ldots\ b_{Nj} \rangle$, where b_{ij} is the ith bit of jth test vector and $b_{ij} \in \{0, 1\}$. For the test vector TV_j, at least one bit b_{ij} must be different from some other bit b_{kj}, where k $\neq$ i.

Definition 2: We define $\mathbf{F_B}$ as the set consisting of all single input bridging faults in a given N-input reversible circuit. The single input bridging fault $F_{(i,j)}$ involves the pair of input lines $\{L_i, L_j\}$, where $\{L_i, L_j\} \subseteq$ L, and is denoted by $\{L_i \leftrightarrow L_j\}$. In other words, F_B = $\{F_{(i,j)} \in \{L_i, L_j\} | \{L_i, L_j\} \subseteq L, F_{(i,j)} \Rightarrow \{L_i \leftrightarrow L_j\}\}$.

For a reversible circuit with N input lines, the number of possible single bridging faults in any level will be ${}^{N}C_2$ or $C(N, 2)$. For example, if $N = 3$, the number of single bridging faults will be C(3,2) = 3, corresponding to the pairs $\{L_1, L_2\}$, $\{L_1, L_3\}$ and $\{L_2, L_3\}$. The corresponding fault set will be F_B = $\{F_{(1,2)}, F_{(1,3)}, F_{(2,3)}\}$.

Definition 3: The test vector TV_k is capable of detecting the bridging fault $F_{(i,j)}$ if and only if $b_{ik} \neq b_{jk}$.

For the set of input lines L = $\{L_1, L_2, L_3\}$, the test vector TV_1 = $\langle 0\ 0\ 1 \rangle$ is capable of applying opposite logic values to the pair of input lines $\{L_1, L_3\}$ and $\{L_2, L_3\}$. Thus TV_1 detects the faults $F_{(1,3)}$ and $F_{(2,3)}$.

Definition 4: The test set $\mathbf{TS_S}$ is the minimal complete test set to detect all the single input stuck-at faults.

Let TV_t=($\langle b_{1t}\ b_{2t}\ \ldots b_{Nt} \rangle$) be a test vector in TS_S, where b_{it} is the ith bit of the tth test vector and $b_{it} \in$ {0,1}. If b_{it}=0 then TV_t is capable of detecting stuck-at-1, otherwise, it detects stuck-at-0.

Definition 5: Let $\mathbf{F_S}$ denote the set of all single input stuck-at faults in a given N-input reversible circuit. Here, we are using the notation $SA\text{-}b_i\text{-}L_i$ for single input stuck-at fault, which denotes a stuck-at fault of ith bit b_i at the ith input line L_i. The stuck-at fault, $SA\text{-}b_i\text{-}L_i$ / $SA\text{-}\overline{b_i}\text{-}L_i$ can only be detected when input line L_i is set to $\overline{b_i}$ / b_i. In other words, F_S={$SA\text{-}b_i\text{-}L_i$, $SA\text{-}\overline{b_i}\text{-}L_i$ | $SA\text{-}b_i\text{-}L_i \Rightarrow \overline{b_i}$ and $SA\text{-}\overline{b_i}\text{-}L_i \Rightarrow b_i$ where, $b_i \in L_i$}.

To compute F_S for each input line L_i, b_i can be set to either 0 or 1. For N lines, the number of single input stuck-at faults will be $2N$.

A. Test Generation for Single Input Bridging Fault

In this section, we discuss the proposed method to generate a minimal test set for detecting all single input bridging faults in a given reversible circuit.

As discussed earlier, a test vector is able to detect a single bridging fault between two lines if it sets the two lines to opposite logic values. The proposed method generates a test set where this property is satisfied for every pair of lines and is hence complete. It is also established that the generated test set is minimal.

The basic idea behind the test generation method is explained below. Let us consider the test set $TS_B = \{TV_1, TV_2, \ \ldots, TV_{\lceil \log_2 N \rceil}\}$. Let the first test vector TV_1 consist of alternating 0's and 1's. The second test vector TV_2 consists of an alternate sequence of two consecutive 0's and two consecutive 1's. Similarly, the nth test vector consists of 2^{n-1} consecutive 0's followed by 2^{n-1} consecutive 1's at the N-input lines L_N, where $1 \leq n \leq \lceil \log_2 N \rceil$. This process is continues, till we get the last test vector $TV_{\lceil \log_2 N \rceil}$. This is shown diagrammatically in Figure 2.

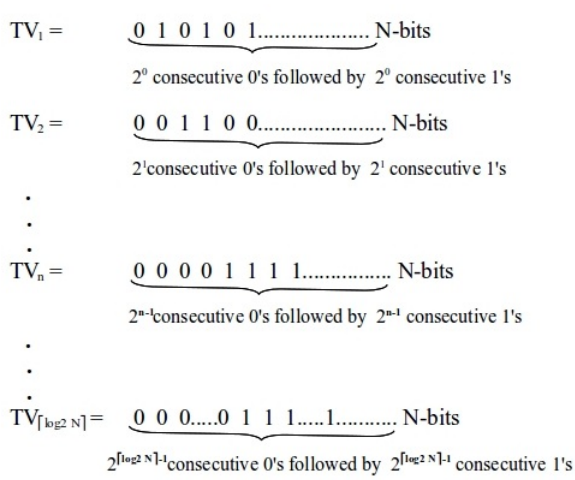

Fig. 2: Test set for detecting single input bridging faults

Example 1: The process of test set generation with the help of examples are shown next. For N=8, the test set TS_B will consist of three test vectors:TV_1=($\langle 0\ 1\ 0\ 1\ 0\ 1\ 0\ 1 \rangle$), TV_2=($\langle 0\ 0\ 1\ 1\ 0\ 0\ 1\ 1 \rangle$), and TV_3=($\langle 0\ 0\ 0\ 0\ 1\ 1\ 1\ 1 \rangle$). The total number of bridging faults at the input is $|F_B| = C(8,2) = 28$. We observe that TV_1 detects the faults $F_{(i,j)}$, where $|j-i| = 1,3,5,7$. Similarly, TV_2 detects the faults $F_{(i,j)}$, where $|j-i| = 2,3,6,7$. Finally, TV_3 detects the faults $F_{(i,j)}$, where $|j-i| = 4,5,6,7$.

For $N = 9$, TS_B will have four test vectors as $\lceil \log_2 9 \rceil = 4$. The four test vectors are, TV_1=($\langle 0\ 1\ 0\ 1\ 0\ 1\ 0\ 1\ 0 \rangle$), TV_2=($\langle 0\ 0\ 1\ 1\ 0\ 0\ 1\ 1\ 0 \rangle$), TV_3=($\langle 0\ 0\ 0\ 0\ 1\ 1\ 1\ 1\ 0 \rangle$) and TV_4=($\langle 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 1 \rangle$). The complete list of single input bridging faults for $N = 8$ and $N = 9$ of this example is shown in Figure 3.

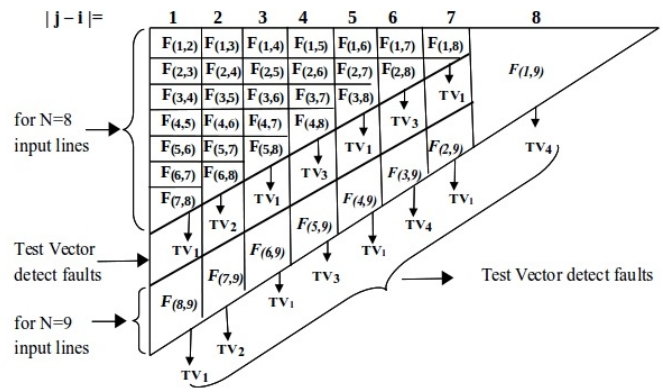

Fig. 3: Demonstration of single input bridging faults for N=8 and N=9

We now present two lemmas with proof that establishes the completeness and minimality of the test set.

Lemma 1: The test set generated using the proposed method detects all single input bridging faults in a given N-input reversible circuit.

Proof: Let us consider the test set for single input bridging faults TS_B={$TV_1, TV_2, \ \ldots, TV_{\lceil \log_2 N \rceil}$}, where the size of any test vector (TV) is N; the number of input line is L_N for the given reversible circuit. The number of test vectors is $\lceil \log_2 N \rceil$. We know that all the single input bridging faults $F_{(i,j)}$ occur for all possible distance $k = |i-j|$, where $1 \leq k \leq N-1$. Total number of test vectors generated by the proposed method is $n = \lceil \log_2 N \rceil$. The nth test vector TV_n=($\langle 0\ 0\ 0\ 0\ \ldots 1\ 1\ 1\ 1\ \ldots \rangle$), contains 2^{n-1} consecutive 0's followed by 2^{n-1} consecutive 1's. By this test vector, input 0 is provided to the first $N/2$ lines and input 1 is provided to the last $N/2$ lines. The distance between first input line L_1 having input 0 and first input line $L_{(N/2)+1}$ having input 1 is $N/2$ and these two input test bits can detect the single input bridging fault $F_{(1,(N/2)+1)}$. By this analogy it is observed that the test vector TV_n can detect all the single input bridging faults $F_{(i,j)}$ where $(N/2) \leq |i-j| \leq N-1$. It is also observed that the fault $F_{((N/2),(N/2)+1)}$ between the input lines $L_{(N/2)}$ and $L_{(N/2)+1}$, which is one distance apart can also be detected by the test vector TV_n. But the fault $F_{(i,i+1)}$ where $1 \leq i \leq (N/2)-1$ cannot be detected by the test vector TV_n. So, the test vector TV_n can detect all possible single input bridging faults $F_{(i,j)}$ where the distance between input lines L_i and L_j lies in the range $[(N/2), \ldots, (N-1)]$. Now consider the $(n-1)^{th}$ test vector TV_{n-1} = ($\langle 0\ 0\ 0\ 0\ \ldots 1\ 1\ 1\ 1\ \ldots 0\ 0\ 0\ 0\ \ldots 1\ 1\ 1\ 1\ \ldots \rangle$), which contains alternate sequence of 2^{n-2} consecutive 0's and 2^{n-2} consecutive 1's. In this test vector the entire test bits

are divided into two parts and the bit pattern of first part is similar to the second part. First part contains the input bits for the lines from L_1 to $L_{(N/2)}$ and the second part contains the input bits for the lines from $L_{(N/2)+1}$ to L_N. With the similar reasoning it can be stated that the test vectors TV_{n-1} can detect all possible single input bridging faults $F_{(i,j)}$ where the distance between input lines L_i and L_j lies in the range $[(N/4), \ldots, ((N/2)-1)]$. Similarly the test vectors TV_2 can detect all possible single input bridging faults $F_{(i,j)}$ where the distance between input lines L_i and L_j lies in the range $[2, 3]$ and test vector T_1 can detect all the single input bridging faults between adjacent lines. It is also observed that the test vector T_1 can detect all faults $F_{(i,j)}$ where $|i-j|$ is an odd number. So, all faults are covered by some test vectors. Therefore, the test set TS_B is the complete test set for detecting all the single input bridging faults in a given reversible circuit.

Lemma 2: The test set TS_B is a necessary minimal test set for detecting all single input bridging faults in a given N-input reversible circuit.

Proof: In Lemma 1, it is established that test set TS_B can detect all single input bridging faults of a given reversible circuit. The number of test vectors in TS_B is $\lceil \log_2 N \rceil$. Consider the kth test vector TV_k that contains alternate sequence of 2^{k-1} consecutive 0's and 2^{k-1} consecutive 1's. This test vector can detect all the faults $F_{(i,j)}$, where the distance between input lines L_i and L_j lies in the range $[(N/2^{n-k+1}), \ldots, (N/2^{n-k}-1)]$. It is observed that every test vector TV_i of TS_B can detect all the faults $F_{(i,j)}$ where the distant between lines L_i and L_j belongs to a binary division of the entire range. So, if we remove any one test vector, then that binary division cannot be covered by the rest of the test vectors of TS_B and bound to miss some single input bridging faults. Also if we replace the test vector TV_i by any other random test vector then some of the faults between some pairs of the input lines whose distance lies in the range $[(N/2^{n-k+1}) \ldots (N/2^{n-k}-1)]$ are bound to be missed. So, the test vector TV_k must be present in the test set TS_B, where $1 \leq k \leq \lceil \log_2 N \rceil$.

Example 2: Consider the reversible benchmark circuit 3_17tc_tfc consisting of six gates as shown in Figure 4. The number of single input bridging faults is $|F_B| = C(3,2) = 3$, and the required number of test vectors is 2. The two test vectors are: TV_1=($\langle 0\ 1\ 0 \rangle$) and TV_2=($\langle 0\ 0\ 1 \rangle$). Table I shows that TS_B is indeed capable of detecting all input bridging faults in the circuit.

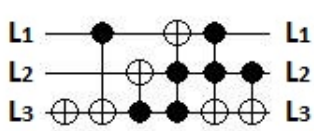

Fig. 4: 3_17tc_tfc benchmark circuit

Example 3: Similarly, the reversible benchmark $4b15g_2$ consists of 15 gates as shown in Figure 5. The number of single input bridging faults for this circuit is $|F_B| = C(4,2) = 6$. Two test vectors are generated as: TV_1=($\langle 0\ 1\ 0\ 1 \rangle$) and TV_2=($\langle 0\ 0\ 1\ 1 \rangle$). Table II shows that the test set detects all single input bridging faults in the circuit.

TABLE I: Bridging fault coverage for 3_17tc_tfc

Single input bridging Fault $F_{(i,j)}$	Test set TS_B	Fault free output	Output for AND-bridging fault	Output for OR-bridging fault
	$L_1\ L_2\ L_3$	$L_1\ L_2\ L_3$	$L_1\ L_2\ L_3$	$L_1\ L_2\ L_3$
$F_{(1,2)}$	0 1 0	0 0 1	1 1 1	1 1 0
$F_{(1,3)}$	0 0 1	0 0 0	1 1 1	0 1 0
$F_{(2,3)}$	0 1 0	0 0 1	1 1 1	0 1 1

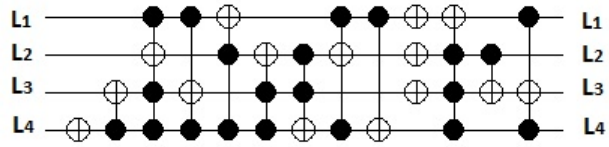

Fig. 5: $4b15g_2$ benchmark reversible circuit

B. Test Set Generation for Single Input Stuck-at Fault

We shall now show how a test set for detecting single stuck-at faults can be derived from the bridging fault test set as generated using the method explained in the previous subsection. We note that if a test set applies opposite logic values 0 and 1 to each of the input lines, then it can detect all single input stuck-at faults. The test set TS_B as discussed in the previous subsection may not satisfy this property for all cases, and hence it is unable to guarantee the detection of all single stuck-at faults.

Here we discuss how we can augment TS_B to generate a test set TS_S that can detect all single input stuck-at faults. Actually, we add one additional test vector to get the required test set as TS_S=$\{TV_1, TV_2, \ \ldots, TV_{\lceil \log_2 N \rceil}, TV_{\lceil \log_2 N \rceil+1}\}$. The extra test vector is generated as $TV_{\lceil \log_2 N \rceil+1}$=($\langle b_{1q}\ b_{2q}\ \ldots\ b_{Nq} \rangle$), where the first bit b_{1q}=1 and the last bit b_Nq=0, while the remaining bits position can be arbitrarily filled up with 0 or 1. This is required because the first bit b_{1i} of the ith ($1 \leq i \leq \lceil \log_2 N \rceil$) test vector in TS_S is always 0, while the last bit b_{ni} of the ith ($1 \leq i \leq \lceil \log_2 N \rceil$) test vector is always 1.

Example 4: For $N = 8$, the test set TS_S will contain four test vectors as: TV_1=($\langle 0\ 1\ 0\ 1\ 0\ 1\ 0\ 1 \rangle$), TV_2=($\langle 0\ 0\ 1\ 1\ 0\ 0\ 1\ 1 \rangle$), TV_3=($\langle 0\ 0\ 0\ 0\ 1\ 1\ 1\ 1 \rangle$), and last test vector TV_4=($\langle 1\ 0\ 0\ 0\ 1\ 1\ 1\ 0 \rangle$). The total number of single input stuck-at faults is $|F_S|$=16. It is observed that the fault SA-0-L_1 cannot be detected by any of the test vectors of TS_B. Similarly, the fault SA-1-L_8 cannot be detected by the test vectors of TS_B. So, it is essential to include the test

TABLE II: Bridging fault coverage for $4b15g_2$

Single input bridging Fault $F_{(i,j)}$	Test set TS_B	Fault free output	Output for AND-bridging fault	Output for OR-bridging fault
	$L_1\ L_2\ L_3\ L_4$	$L_1\ L_2\ L_3\ L_4$	$L_1\ L_2\ L_3\ L_4$	$L_1\ L_2\ L_3\ L_4$
$F_{(1,2)}$	0 1 0 1	1 0 1 0	1 1 0 0	0 0 1 1
$F_{(1,3)}$	0 0 1 1	1 1 1 0	1 1 0 0	0 1 1 1
$F_{(1,4)}$	0 0 1 1	1 1 1 0	0 1 0 1	0 1 1 1
$F_{(2,3)}$	0 1 0 1	1 0 1 0	1 1 0 0	1 0 1 1
$F_{(2,4)}$	0 0 1 1	1 1 1 0	0 1 0 1	1 0 1 1
$F_{(3,4)}$	0 1 0 1	1 0 1 0	0 0 0 0	1 0 1 1

vector TV_4 in the test set TS_S, which can detect both the faults SA-0-L_1 and SA-1-L_8.

The following two results are established for the proposed algorithms for minimality and completeness.

Lemma 3: The test set TS_S detects all single input bridging faults and single input stuck-at faults in a given reversible circuit.

Proof: In Lemma 1, it is established that all the single input bridging faults are detected by the test set generated by the proposed method. Let us consider the test set TS_S=$\{TV_1, TV_2, \ldots, TV_p, TV_q\}$, where $p = \lceil \log_2 N \rceil$, $q = \lceil \log_2 N \rceil + 1$. It is observed that the test vectors TV_1 to TV_p apply 0 for input line L_1. Similarly, 0 is applied to input line L_2 with respect to test vector TV_1 to TV_{p-1}, and 1 is applied to L_2 by the test vector TV_p. Similarly, a 1 is applied to L_N with respect to test vectors TV_1 to TV_p. Due to this binary encoding pattern, the first line L_1 always gets a 0 as input and the last line L_N gets a 1 as input. Due to this reason these test vectors are capable of detecting all the single input faults SA-0-L_i and SA-1-L_i except the faults SA-0-L_1 and SA-1-L_N. However, the fault SA-1-L_N cannot be detected by the test set TS_B if $N = 2^x$ for some integer $x \geq 2$. Due to the inclusion of the test vector TV_q in the test set TS_S, it is possible to detect the faults SA-0-L_1 and SA-1-L_N. Hence, we can conclude that TS_S detects all single input stuck-at as well as all single input bridging faults.

Lemma 4: TS_S is a minimal test set for detecting all single input bridging faults and single input stuck-at faults in a given reversible circuit.

Proof: In Lemma 2, it is established that all the test vectors of the test set TS_B are essential to detect all single input bridging faults in the circuit. Also it is shown that the test set TS_B is minimal. In Lemma 3, it is proved that the test set TS_S=$\{TV_1, TV_2, \ldots, TV_p, TV_q\}$ is the complete set of test vectors for detecting all single input stuck-at faults. Since TV_1 to TV_p are essential for detecting single input bridging faults, none of these can be removed from TS_S. Since all the single input stuck-at faults cannot be detected by the test set TS_B, so, requirement of additional test vector is unavoidable. Since only one additional test vector has been added to the test set, so it establishes the minimality of the test set TS_S. Hence, the test set TS_S is a necessary and minimal test set for detecting all the single input bridging faults and single input stuck-at faults for a given reversible circuit.

Example 5: Consider the reversible benchmark circuit 3_17tc_tfc as shown in Figure 4. The number of single input stuck-at faults in this circuit is $|F_S|$=6. According to the proposed method for generating the test set, the generated test set is TS_S=$\{TV_1, TV_2, TV_3\}$, where TV_1=($\langle 0\ 1\ 0 \rangle$), TV_2=($\langle 0\ 0\ 1 \rangle$) and TV_3=($\langle 1\ 0\ 0 \rangle$). Table III shows that the test set TS_S can detect all the single input stuck-at faults in the circuit.

Example 6: Consider the reversible benchmark circuit $4b15g_2$ which is shown in Figure 5. The number of single input stuck-at faults of this circuit is $|F_S|$=8. According to the proposed test set generation method, the test set is TS_S=$\{TV_1, TV_2, TV_3\}$, where TV_1=($\langle 0\ 1\ 0\ 1 \rangle$), TV_2=($\langle 0\ 0\ 1\ 1 \rangle$) and TV_3=($\langle 1\ 0\ 1\ 0 \rangle$). Table IV shows that the test set TS_S is capable of detecting all the single input stuck-at faults in the circuit.

TABLE III: Stuck-at fault coverage for 3_17tc_tfc

Single input stuck-at Fault SA-b_i-L_i	**Test set** TS_S	**Fault free output**	**Output for** SA-0-L_i	**Output for** SA-1-L_i
	$L_1\ L_2\ L_3$	$L_1\ L_2\ L_3$	$L_1\ L_2\ L_3$	$L_1\ L_2\ L_3$
SA-0-L_1	1 0 0	1 0 0	1 1 1	not required
SA-1-L_1	0 1 0	0 0 1	not required	1 1 0
SA-0-L_2	0 1 0	0 0 1	1 1 1	not required
SA-1-L_2	0 0 1	0 0 0	not required	0 1 1
SA-0-L_3	0 0 1	0 0 0	1 1 1	not required
SA-1-L_3	1 0 0	1 0 0	not required	0 1 0

TABLE IV: Stuck-at fault coverage for $4b15g_2$

Single input stuck-at Fault SA-b_i-L_i	**Test set** TS_S	**Fault free output**	**Output for** SA-0-L_i	**Output for** SA-1-L_i
	$L_1\ L_2\ L_3\ L_4$	$L_1\ L_2\ L_3\ L_4$	$L_1\ L_2\ L_3\ L_4$	$L_1\ L_2\ L_3\ L_4$
SA-0-L_1	1 0 1 0	0 0 0 1	0 1 0 1	not required
SA-1-L_1	0 1 0 1	1 0 1 0	not required	0 0 1 1
SA-0-L_2	0 1 0 1	1 0 1 0	1 1 0 0	not required
SA-1-L_2	0 0 1 1	1 1 1 0	not required	1 0 1 1
SA-0-L_3	0 0 1 1	1 1 1 0	1 1 0 0	not required
SA-1-L_3	0 1 0 1	1 0 1 0	not required	1 0 1 1
SA-0-L_4	0 0 1 1	1 1 1 0	0 1 0 1	not required
SA-1-L_4	1 0 1 0	0 0 0 1	not required	0 1 1 1

IV. Experimental results

The algorithms for the proposed test generation approach have been implemented and evaluated on benchmarks using to verify their correctness. Several reversible benchmark circuits, based on the NCT and GT gate libraries, have been considered [23]. We have experimented with circuits with up to 40 lines. The number of single input bridging faults in a circuit with 40 lines is more than 20,000, and our proposed approach is able to detect all the faults. We have compared our result with a previous work [22]. The column 8 in Table V indicates the differences in the number of test vectors generated by the method of [22] and our proposed method. It is observed that the number of test vectors generated by our method is always less than that by the existing method, and still able to provide 100% fault coverage.

V. Conclusion and Future work

This paper discusses a scheme for minimal test generation to detect all input stuck-at and bridging faults in reversible circuits. First it has been proved formally that $\lceil \log_2 N \rceil$ number of test vectors are necessary and sufficient for detection of all the bridging faults of a reversible circuit with N inputs. Then it is illustrated that just another additional vector is sufficient to cover all the stuck at faults. Experimental results showed that the test size of the proposed scheme is much smaller as compared to existing methods yet maintaining 100% fault coverage.

As a future work, the same principle may be attempted to cover other fault models.

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TABLE V: Detection of input bridging and stuck-at faults (S = Stuck-at faults and B = Bridging faults)

Benchmark Circuit	Gate Model	No. of Input/ Output	No.of input stuck-at faults	No. of input bridging fault	No. of test vectors [22]	No. of test vectors [Proposed]	Difference between [22] & [Proposed]	% Fault Coverage [Proposed]
					B+S	B+S	B+S	B+S
ham3	NCT	3/3	6	8	-	3	—	100
3_17	NCT	3/3	6	8	3	3	**0**	100
nth_ prime3 _inc	NCT	3/3	6	8	—	3	–	100
4b15g_2	NCT	4/4	8	22	—	3	–	100
hwb4	NCT	4/4	8	22	—	3	–	100
hwb4	GT	4/4	8	22	—	3	–	100
2of5	NCT	5/1	10	48	—	4	—	100
Xor5	NCT	5/1	10	48	—	4	—	100
6sym	NCT	6/1	12	90	6	4	**2**	100
hwb6	GT	6/6	12	90	6	4	**2**	100
hwb7	GT	7/7	14	152	7	4	**3**	100
rd73	NCT	7/3	14	152	7	4	**3**	100
ham7	GT	7/7	14	152	7	4	**3**	100
rd84	NCT	8/4	16	238	8	4	**4**	100
9sym	NCT	9/1	18	352	9	5	**4**	100
ham15	GT	15/15	30	1848	15	5	**10**	100
mod 1024	GT	20/20	40	4598	20	6	**14**	100
mod 1048576 adder	GT	40/40	80	21222	40	7	**33**	100

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